

MGTA 495 Fraud Analytics

Spring 2026

CLASS TIMES: Thursdays 2 to 4:50 pm

ROOM: None. All classes online

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OFFICE HOURS: by request

Instruction begins on Thursday, April 2, 2026, and the last class is June 4.

DESCRIPTION

In this course you will learn everything you need to know in order to build statistical/machine learning models to find fraud in data events. We will cover the basics of building fraud models: understanding the problem dynamics, modes of fraud, data analysis and preparation, building expert variables/features, different algorithm choices, and measurements of model goodness. There will be ten individual homework problems that go through the entire process from raw data to finished fraud algorithm. There will be two projects, one a classic supervised problem on credit card transaction data and the other an unsupervised fraud algorithm on a NY property data set.

We will talk through in detail several examples fraud problems and solutions, including tax fraud, application identity fraud, healthcare claims fraud, and credit card transaction fraud. We will cover the topic of statistical modeling with a drill-down into all steps to building machine learning models. There will be a high-level discussion of a wide variety of statistical modeling methods, including linear/logistic regressions, clustering, neural nets, decision trees, random forests, deep learning, boosting and bagging. The overall fraud solution emphasis will be on creating expert variables/features which are the key to building successful fraud algorithms.

The lectures will go into detail on how to build both supervised and unsupervised fraud models. When we have records that are known to be fraudulent (labeled as fraud) then we can usually build supervised models to predict new fraud instances going forward. In many cases we don't have labeled fraud records, meaning we don't have any or sufficient specific examples of fraud events from the past. In this case we typically build unsupervised fraud models, where we look for anomalous records that are unusual in some important way.

Other topics covered will be forensic accounting vs. real-time fraud algorithms, encoding of difficult fields, imputing missing data values, encoding categorical variables, variable scaling, fuzzy matching/linking, and preparing Data Quality Reports.

The three projects to be completed are (1) building a real-time credit card transaction fraud scoring model, (2) finding anomalies/potential tax fraud in a large, messy data set of NY properties, and (3) application identity fraud. The second project is an unsupervised forensic accounting type problem, and the first and third projects will be supervised, where we can evaluate the model performance and estimate the business impact. The data sets are real and come from an open datasets (1) credit card transactions, (2) NY property tax records and (3) data synthesized from applications for loans and cell phones. For each of the projects the students will describe and execute the fraud algorithms and prepare a business reports.

Students are expected to have good backgrounds in math and statistics, and a working ability in Python. Many assignments will require the use of Python programming.

OBJECTIVES

After a successful completion of this course you will be able to

- Understand the basics of any business fraud problem
- Build a new fraud algorithmic solution to any practical, real world problem
- Participate in a conversation on any state-of-the-art fraud algorithmic solution
- Describe in detail how to handle incomplete data, including encoding complex fields
- Build professional business reports on a complex data algorithmic topic

MATERIALS

Required

Machine access for doing statistical data analysis, coding in Python, access to open source machine learning libraries.

Suggested books:

- *Fraud Analytics Using Descriptive, Predictive, and Social Network Techniques*, Baesens, Van Vlasselaer, Verbeke.
- *Bank Fraud*, Subramanian.
- *Foundations of Predictive Analytics*, Wu, Coggeshall

SCHEDULE

ASSIGNMENTS

There are ten homework assignments to be done individually. It's fine to collaborate on any or all the work, but it's best for each student to complete each homework as much as possible individually.

Reading Assignments

None assigned at this time but some material may be suggested and provided, such as the HNC Falcon patent.

Class Projects

You will perform three projects on real-world data sets to build working fraud algorithms. The projects will be completed individually through a series of homework problems.

In the projects you will learn to analyze and clean a large, dirty real data set, and then build a Data Quality Report describing the data. There will be missing fields that need to be filled in with various methodologies. You will then build a supervised fraud algorithm on this data, using a proper pipeline of methodologies: separation of data into training/testing/validation sets, expert variable creation, feature selection, using a supervised machine learning algorithm, and evaluating the results with various measures of goodness.

In the second project you will be performing a forensic accounting-type analysis looking for past anomalies that are candidates for possible fraud transactions.

Exams

There will be no exams for this course.

Class Participation

There will be periodic open discussions during the class hours and students are expected to participate in these conversations.

GRADING

Points will be assigned as shown in the table below. A distribution of student total points will be used to assign individual grades so about half the students get As and half get Bs or lower.

Assignments	Points
10 individual homeworks: 7 of the homeworks are worth 10 points and 3 (the reports) are worth 20 points each.	130
Class participation 10 points	10
Total points	140

The total of points for the class assignments is 140, and the final grade will be determined by the student's location in the class score distribution. In general, half the students will get B's and half A's.

COURSE POLICIES

Homework and projects are due at midnight on the day before the next class, or as otherwise assigned. Points will be deducted for late submissions.

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of university intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://www-senate.ucsd.edu/manual/appendices/app2.htm#AP14>

How the Honor Code applies to this course:

This course will be preparing you for professional positions in industry. There you will be expected to be able to work in teams, where complex projects are divided into tasks to be completed by individual members. In this environment every person's contribution is expected and explicitly known. In your team projects it is expected that each member contributes substantively to the final product, as is expected in an actual work situation.

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.